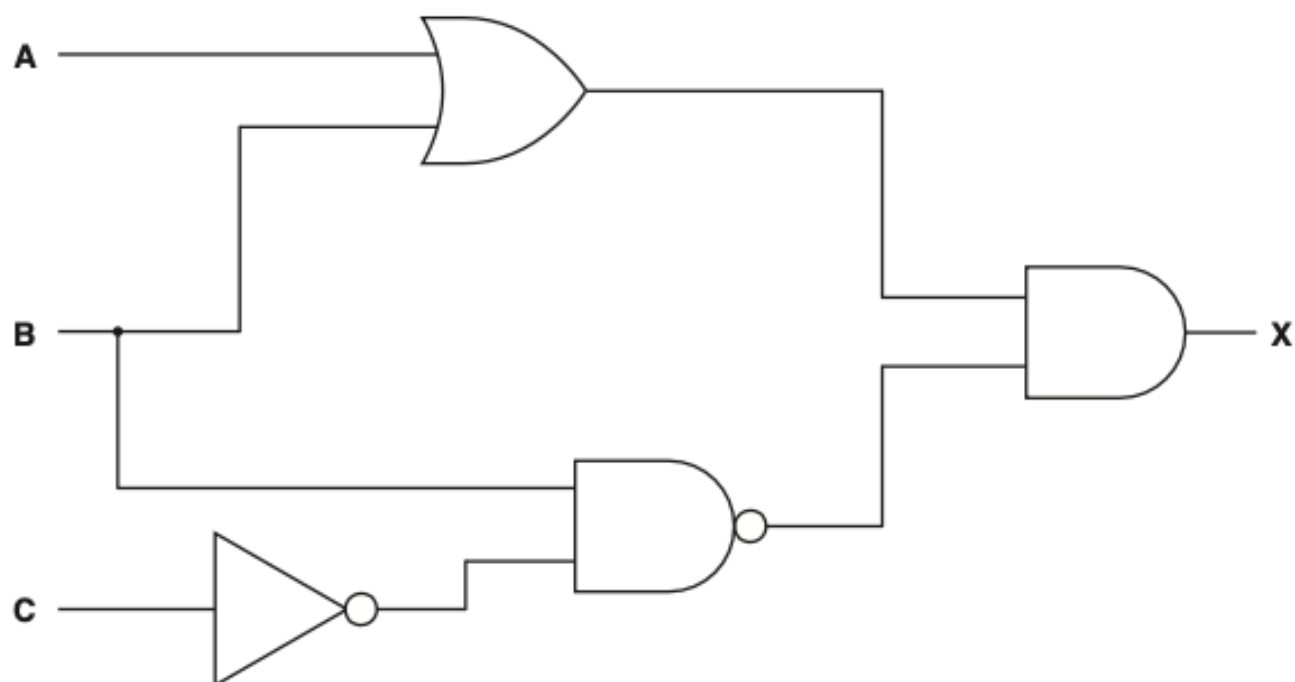


- 6 (a) Complete the truth table for the given logic circuit.

Do **not** attempt to simplify the logic circuit.



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

- 6 (b) A water control system uses a switch and two pressure sensors.

The outputs of the switch and sensors are shown in the table.

Sensor or Switch	Output of 1	Output of 0
Switch (S1)	On	Off
Pressure Sensor (P1)	≥ 3	< 3
Pressure Sensor (P2)	≥ 3	< 3

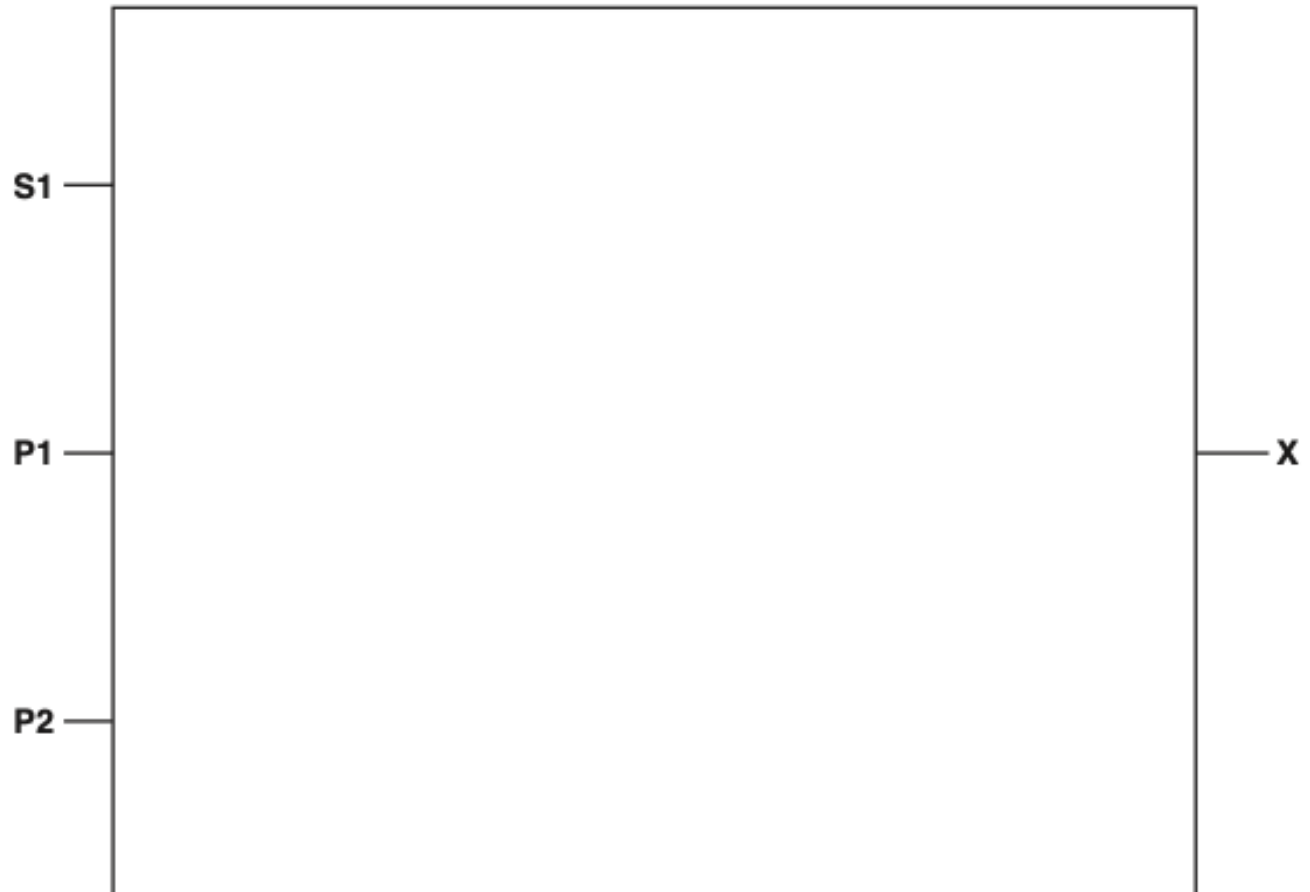
Create a logic circuit that will produce an output (X) of 1 when:

The switch **S1** is on

and

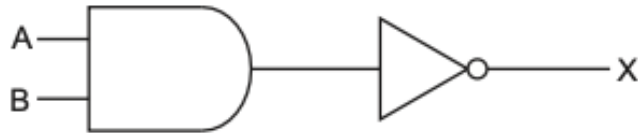
either P1 is less than 3 or P2 is less than 3, but not both.

All logic gates used must have a maximum of two inputs.



4(a) Identify the name **and** draw the **single** logic gate that can replace the given logic circuits.

(i)



Name of gate:

Drawing of gate:

[2]

(ii)



Name of gate:

Drawing of gate:

[2]

(b) Complete the truth table for the given logic statement:

$$X = (((A \text{ OR } C) \text{ AND } (\text{NOT } A \text{ AND } \text{NOT } C)) \text{ XOR } B)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

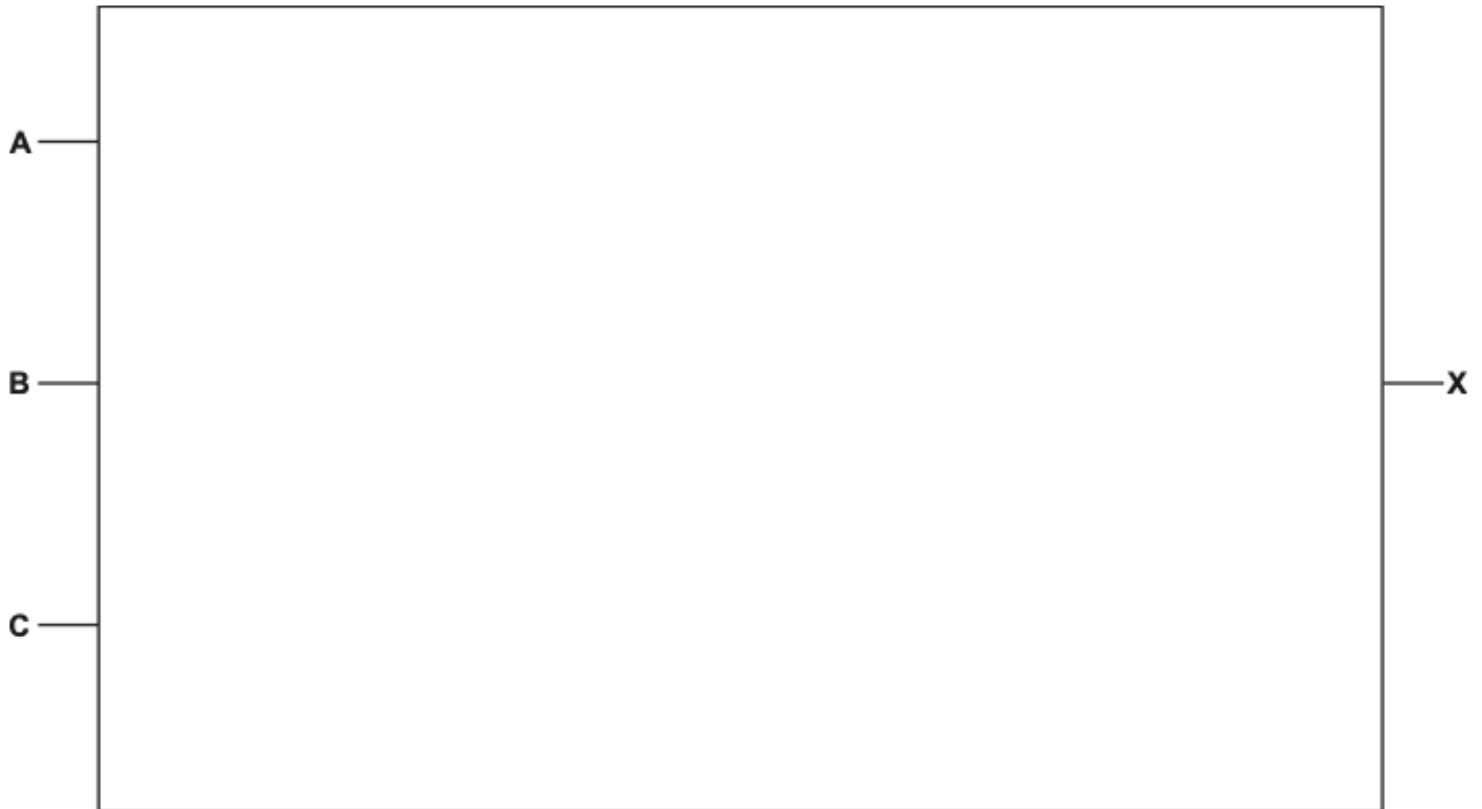
[4]

6 Consider the given logic statement:

$$X = (((A \text{ XOR } B) \text{ AND } C) \text{ OR NOT } C)$$

(a) Draw a logic circuit to match the given logic statement.

All logic gates must have a maximum of **two** inputs. Do **not** attempt to simplify the logic statement.



[4]

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

4 Consider the given logic statement:

$$X = ((\text{NOT } (A \text{ NAND } B)) \text{ OR } (B \text{ NOR } C))$$

(a) Draw a logic circuit to represent the given logic statement.

All logic gates must have a maximum of **two** inputs. Do **not** attempt to simplify the logic statement.



[4]

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

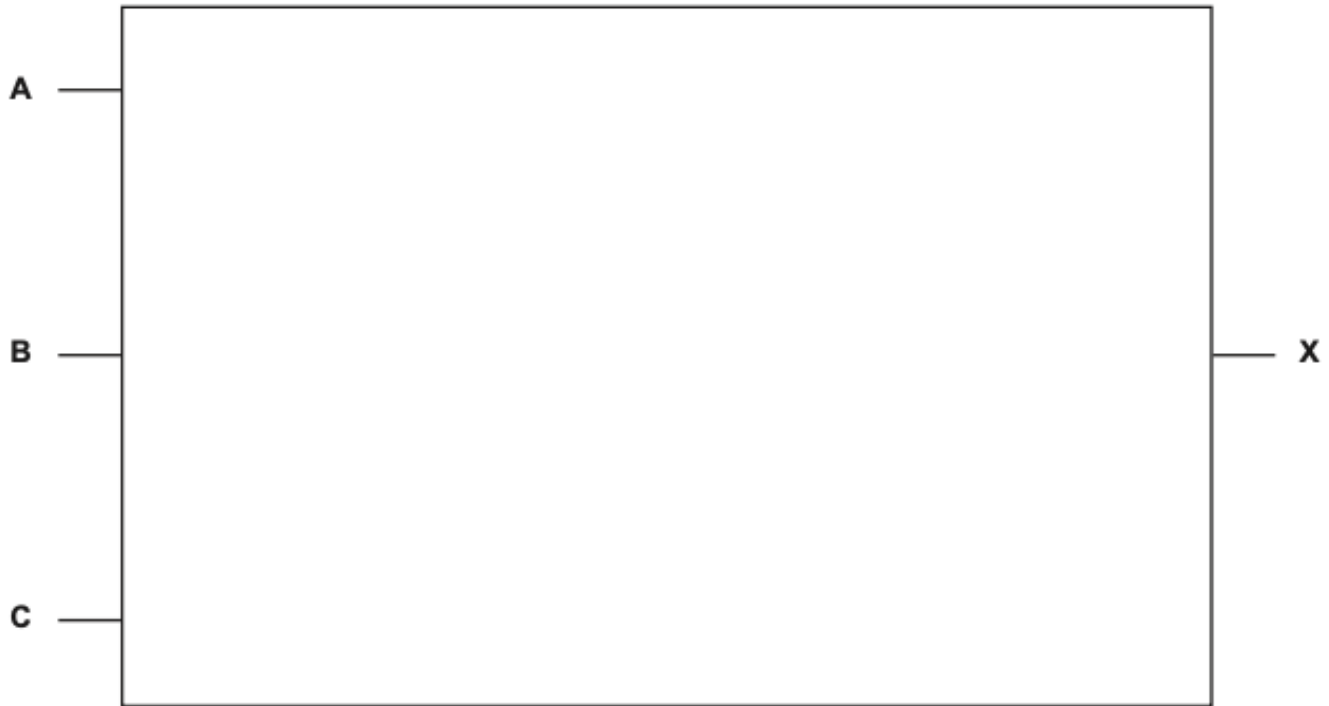
[4]

4 Consider the logic statement:

$$X = (((A \text{ NAND } B) \text{ NOR } (B \text{ AND } C)) \text{ OR } C)$$

(a) Draw a logic circuit to match the given logic statement.

All logic gates must have a maximum of **two** inputs. Do **not** attempt to simplify the logic statement.



[4]

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

10 Consider the following logic statement:

$$X = ((B \text{ AND NOT } A) \text{ XOR } (A \text{ OR } C))$$

(a) Draw a logic circuit to match the given logic statement.

All logic gates must have a maximum of **two** inputs. Do **not** attempt to simplify the logic statement.



[4]

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

5 Computers use logic gates.

(a) State the **single** logic gate that produces each truth table.

Truth table			Logic gate														
<table><tr><th>A</th><th>B</th><th>Output</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Output	0	0	1	0	1	1	1	0	1	1	1	0		
A	B	Output															
0	0	1															
0	1	1															
1	0	1															
1	1	0															
<table><tr><th>A</th><th>B</th><th>Output</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Output	0	0	0	0	1	1	1	0	1	1	1	0		
A	B	Output															
0	0	0															
0	1	1															
1	0	1															
1	1	0															
<table><tr><th>A</th><th>B</th><th>Output</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Output	0	0	1	0	1	0	1	0	0	1	1	0		
A	B	Output															
0	0	1															
0	1	0															
1	0	0															
1	1	0															

- (b) An aeroplane has a warning system that monitors the height of the aeroplane above the ground, whether the aeroplane is ascending or descending, and the speed of the aeroplane.

Input	Binary value	Condition
Height (H)	1	Height is less than 500 metres
	0	Height is greater than or equal to 500 metres
Ascending or Descending (A)	1	Aeroplane is ascending or in level flight
	0	Aeroplane is descending
Speed (S)	1	Speed is less than or equal to 470 knots
	0	Speed is greater than 470 knots

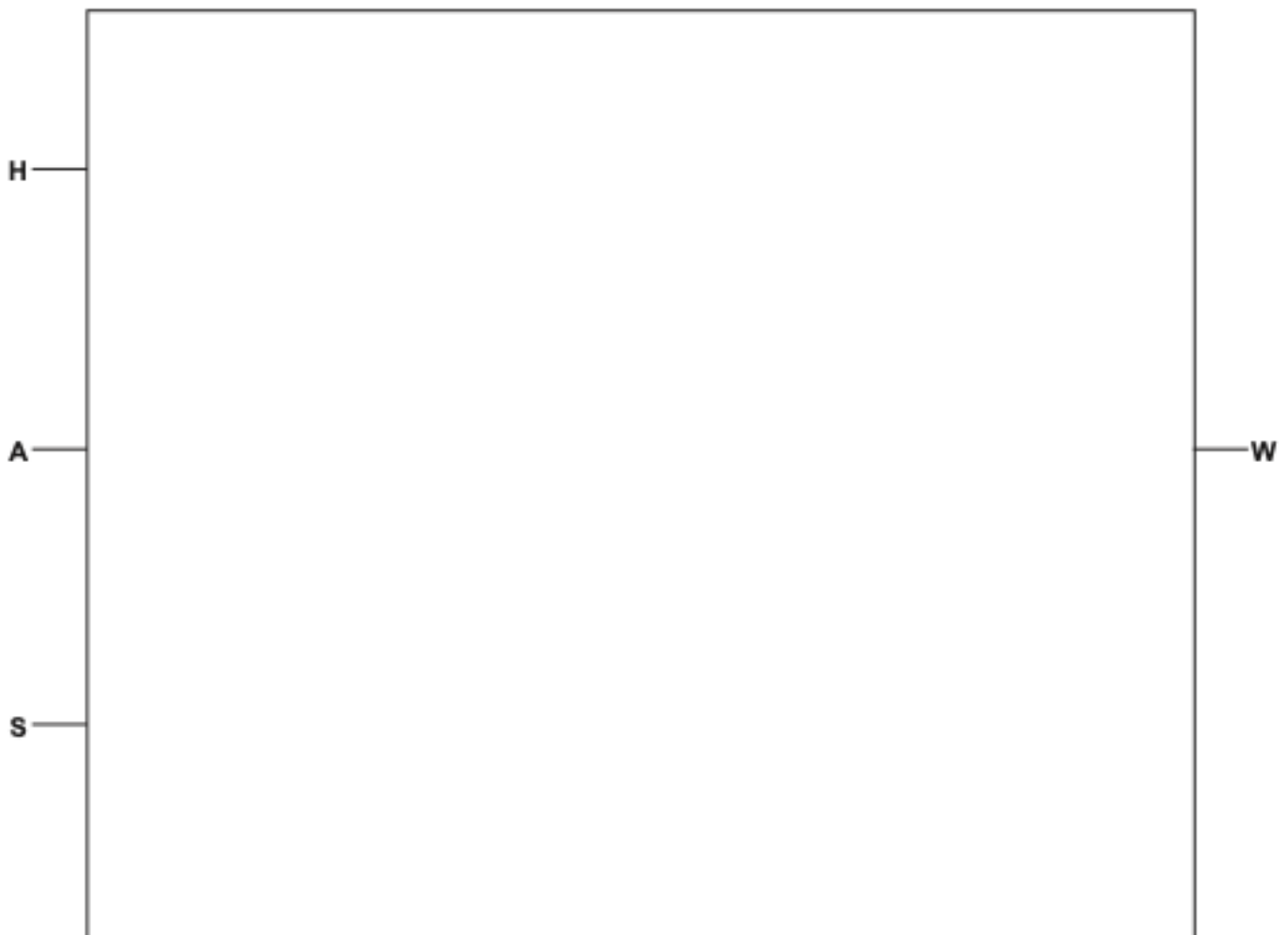
The warning system will produce an output of 1 that will sound an alarm (W) when either of these conditions apply:

Height is less than 500 metres and the aeroplane is descending

or

The aeroplane is descending and speed is greater than 470 knots

Draw a logic circuit to represent the warning system.

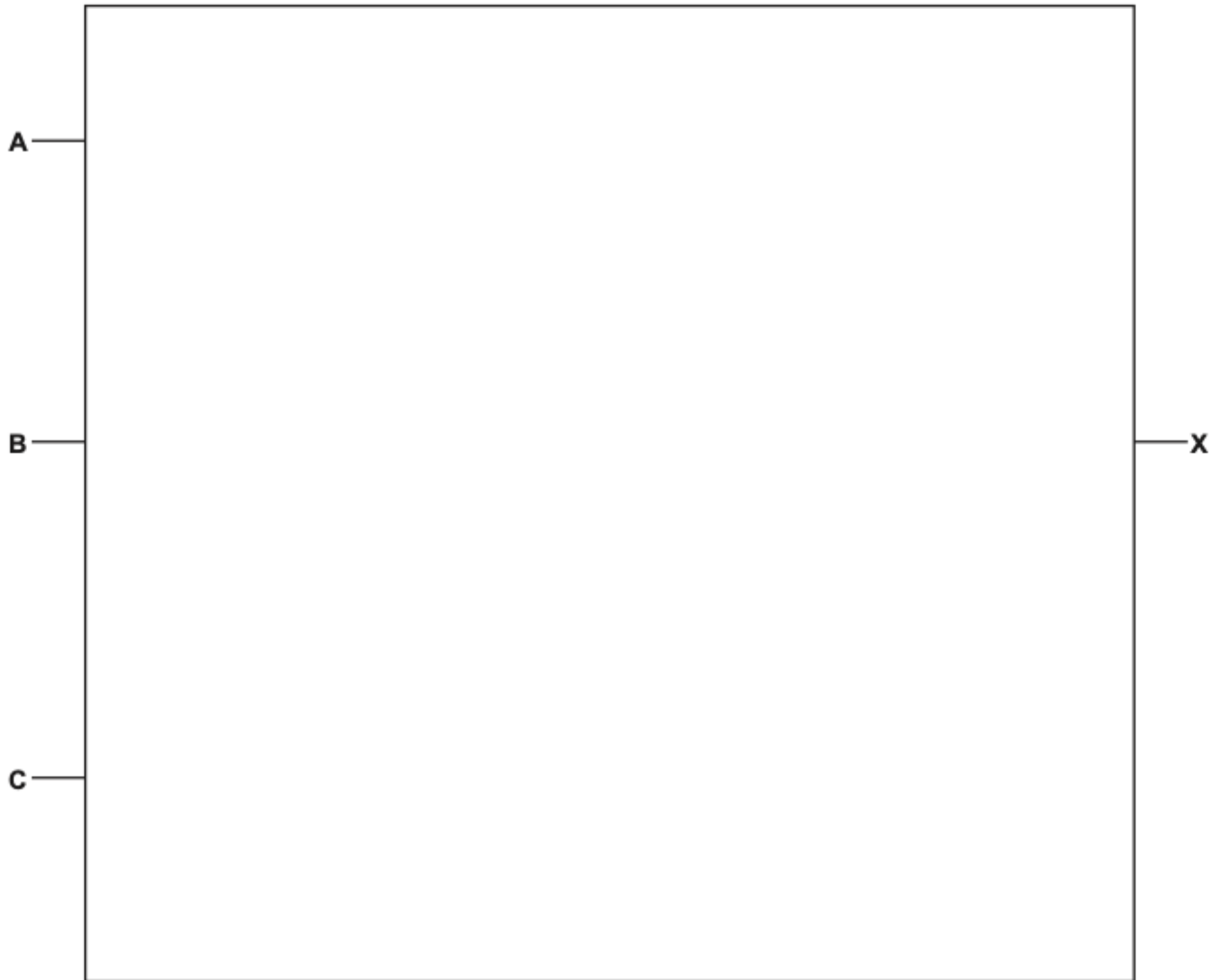


6 Consider the logic statement:

$$X = (((\mathbf{A} \text{ AND } \mathbf{B}) \text{ OR } (\mathbf{C} \text{ AND NOT } \mathbf{B})) \text{ XOR NOT } \mathbf{C})$$

(a) Draw a logic circuit to represent the given logic statement.

Do **not** attempt to simplify the statement. All logic gates must have a maximum of two inputs.



[6]

(b) Consider the completed truth table for the given logic statement.

Row number	A	B	C	Working space	X
1	0	0	0		0
2	0	0	1		1
3	0	1	0		0
4	0	1	1		1
5	1	0	0		0
6	1	0	1		1
7	1	1	0		0
8	1	1	1		1

There are four errors in the truth table in the output (X) column.

Identify the **four** incorrect outputs.

Write the row number to identify each incorrect output.

Row

Row

Row

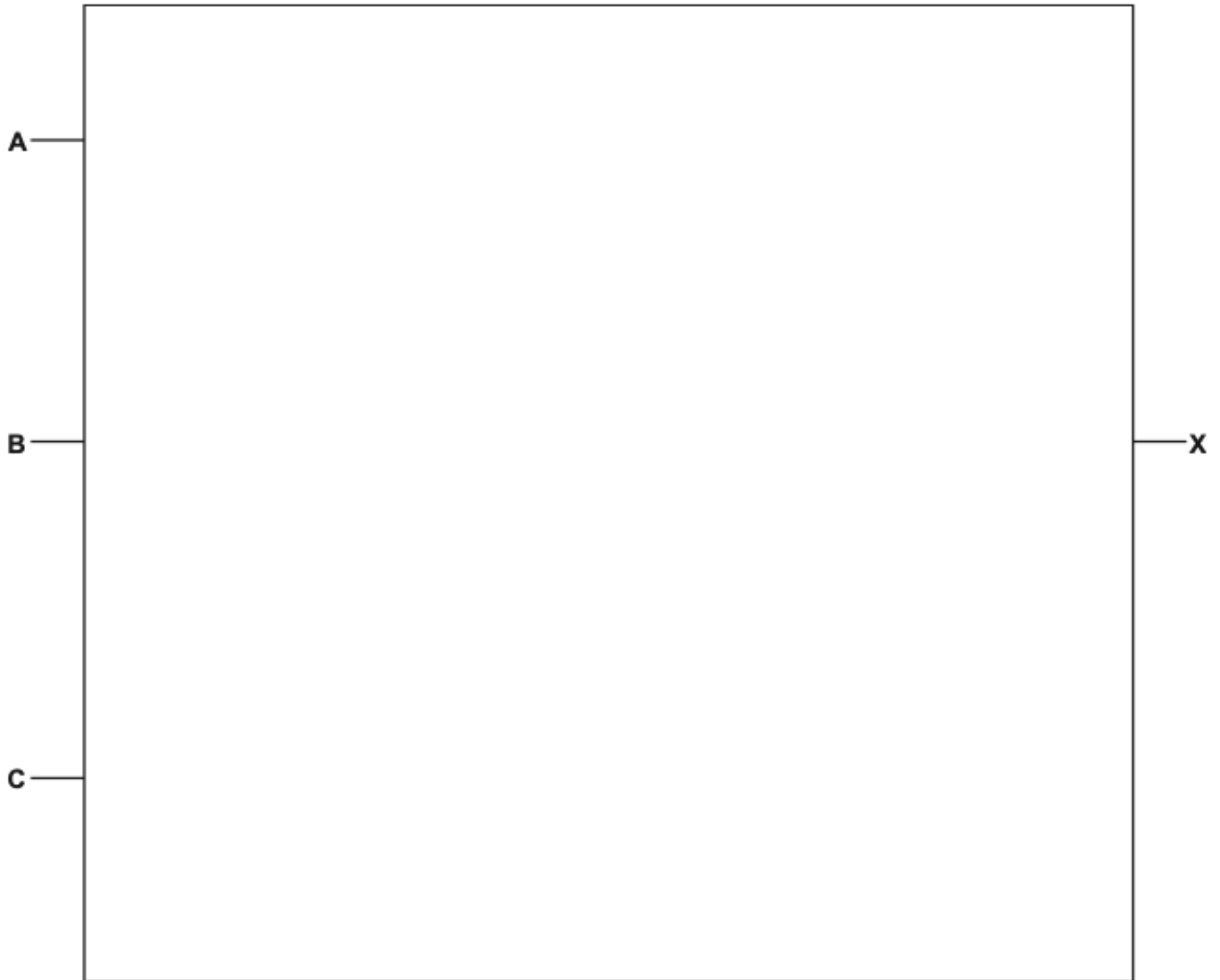
Row

3 Consider the logic statement:

$$X = (((\text{NOT } A \text{ AND } B) \text{ OR } C) \text{ AND } B) \text{ NOR } (B \text{ OR } C))$$

(a) Draw a logic circuit to represent the given logic statement.

Do **not** attempt to simplify the statement. All logic gates must have a maximum of **two** inputs.



[6]

(b) Consider the completed truth table for the given logic statement.

Row number	A	B	C	Working space	X
1	0	0	0		1
2	0	0	1		1
3	0	1	0		1
4	0	1	1		0
5	1	0	0		1
6	1	0	1		0
7	1	1	0		1
8	1	1	1		1

There are four errors in the truth table in the output (X) column.

Identify the **four** incorrect outputs.

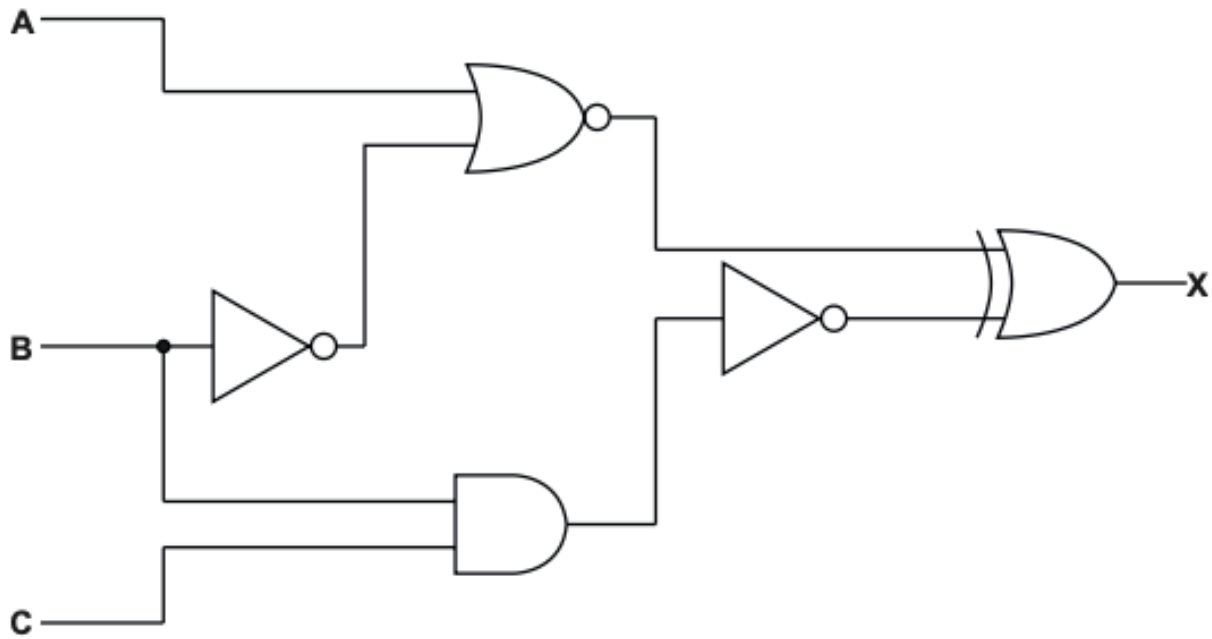
Write the row number to identify each incorrect output.

Row

Row

Row

Row



8 (a) Two **NOT** gates are used in the given logic circuit.

Identify **three** other logic gates that are used in the given logic circuit.

- 1
- 2
- 3

[3]

8 (b) Consider the completed truth table for the given logic circuit.

Row number	A	B	C	Working space	X
1	0	0	0		0
2	0	0	1		1
3	0	1	0		0
4	0	1	1		0
5	1	0	0		1
6	1	0	1		1
7	1	1	0		0
8	1	1	1		1

There are four errors in the truth table in the output (X) column.

Identify the **four** incorrect outputs.

Write the row number to identify each incorrect output.

Row

Row

Row

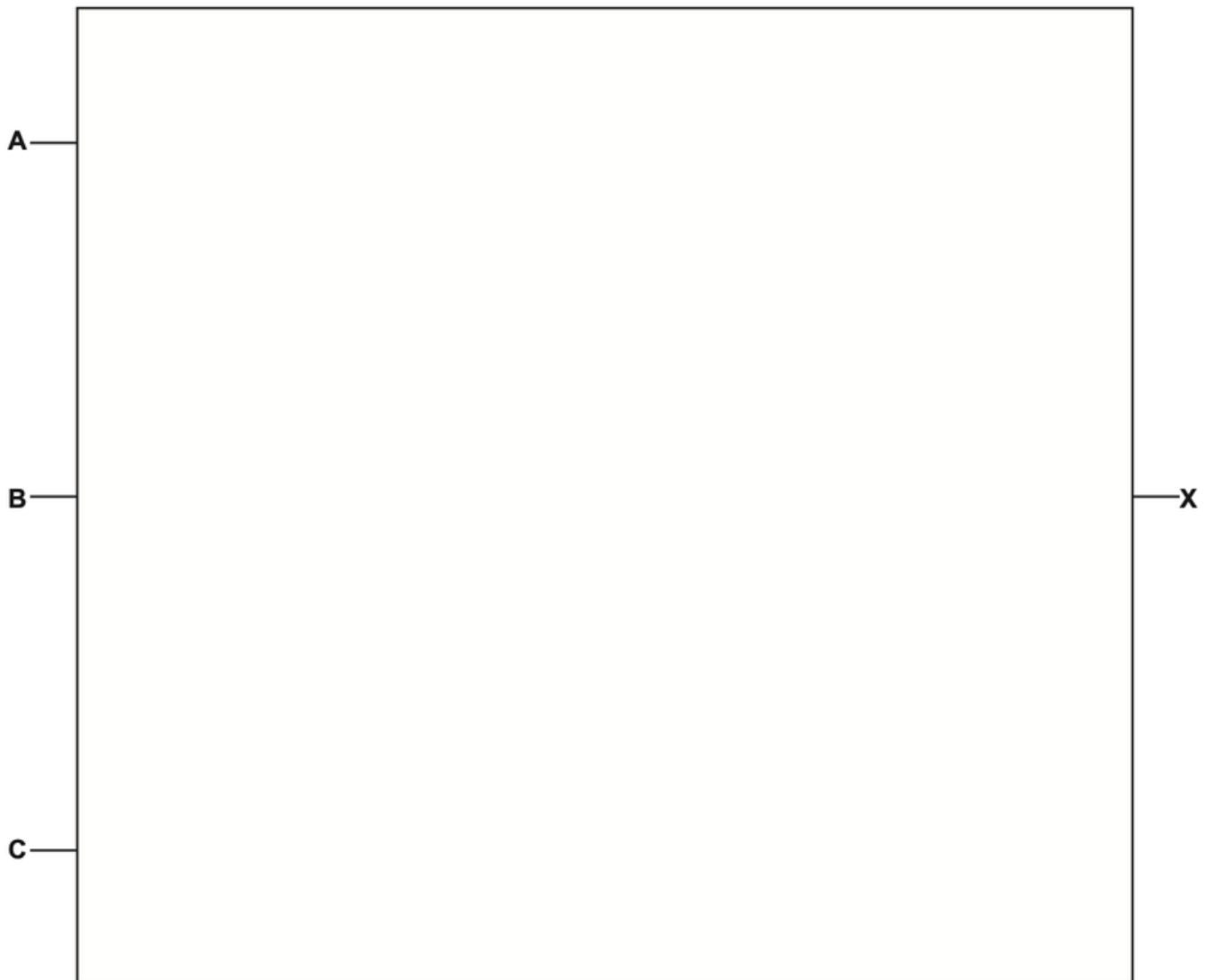
Row

10 Consider the following logic statement:

$$X = (((A \text{ OR } B) \text{ OR } (\text{NOT } (B \text{ XOR } C))) \text{ AND } C)$$

(a) Draw a logic circuit to represent the given logic statement.

Do **not** attempt to simplify the logic statement. All logic gates must have a maximum of **two** inputs.



[5]

(b) State the name of a logic gate that does **not** appear in the logic statement and draw the symbol for the logic gate.

Name of logic gate

Logic gate symbol:



[2]

(c) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

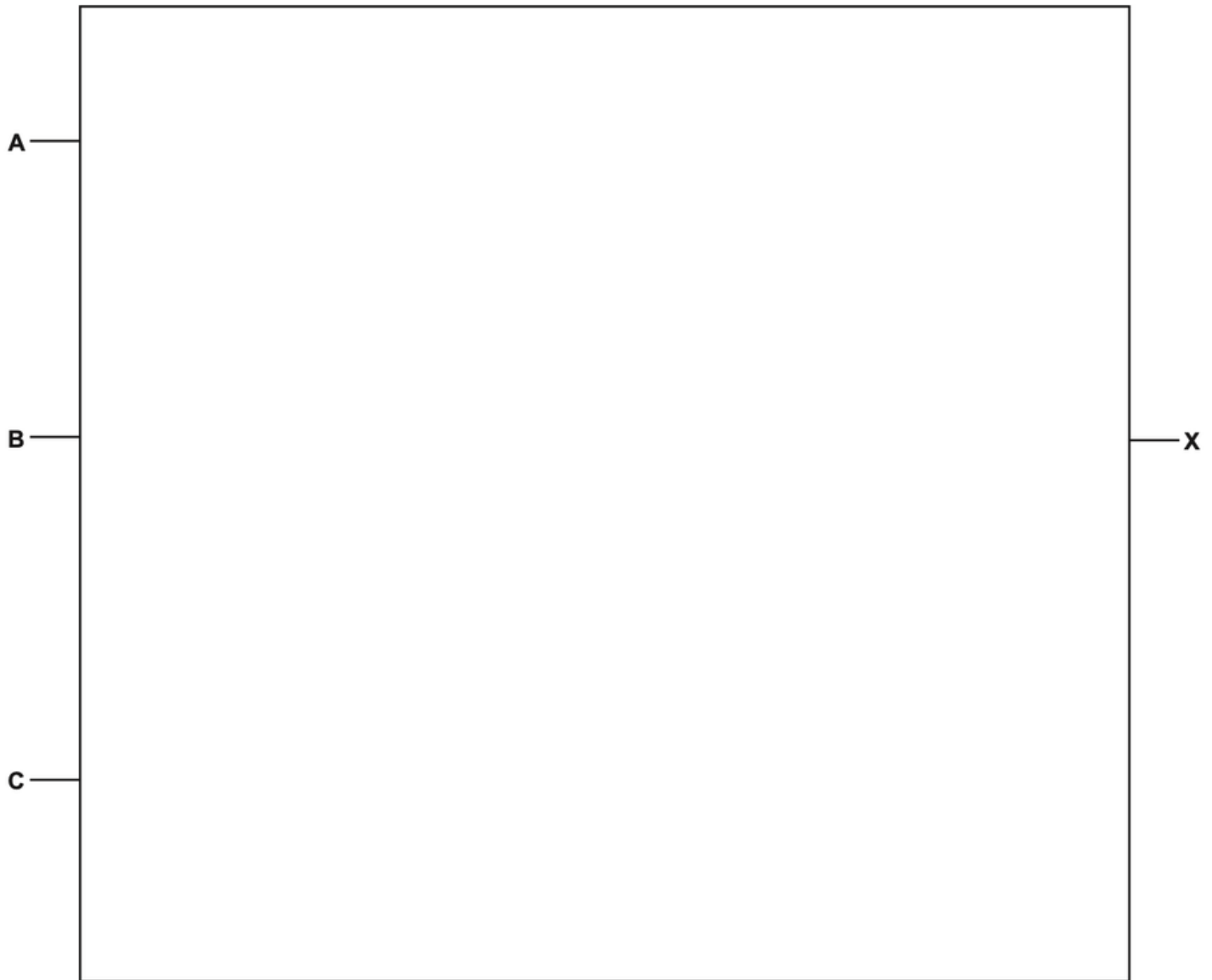
[4]

11 Consider the following logic statement:

$$X = (((A \text{ AND } B) \text{ OR } (\text{NOT } (B \text{ OR } C))) \text{ NAND } C)$$

(a) Draw a logic circuit to represent the given logic statement.

Do **not** attempt to simplify the logic statement. All logic gates must have a maximum of **two** inputs.



[5]

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

(c) Identify **two** logic gates that are **not** included in the given logic statement.

Logic gate 1

Logic gate 2

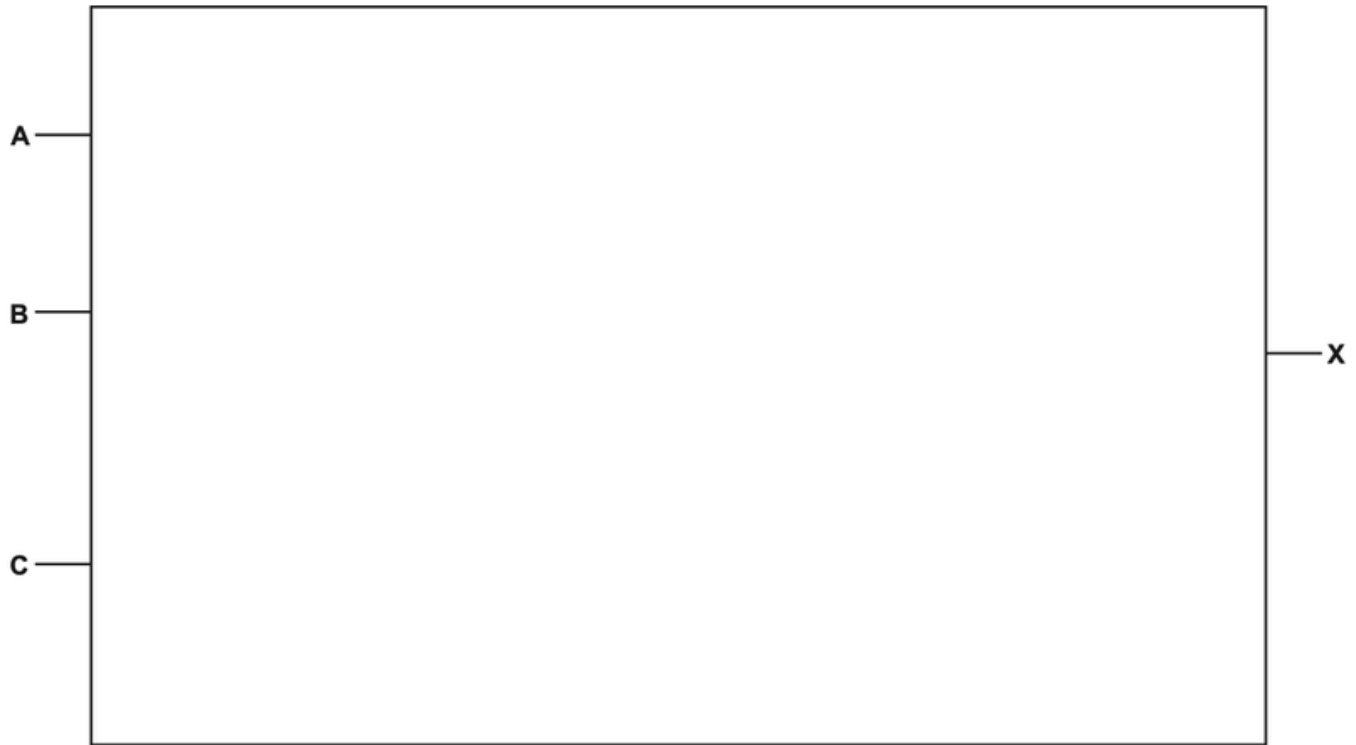
[2]

8 Consider the following logic statement:

$$X = (((A \text{ OR } B) \text{ AND } (\text{NOT}(B \text{ XOR } C))) \text{ OR } \text{NOT } C)$$

(a) Draw a logic circuit to represent the given logic statement.

Do **not** attempt to simplify the logic statement. All logic gates must have a maximum of **two** inputs.



[6]

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

- 3 (a) Tick (✓) to show which logic gates will give an output of 1 for the given inputs A and B.

Inputs	AND	OR	NAND	NOR	XOR
A = 1 B = 1					
A = 0 B = 0					
A = 1 B = 0					

[3]

- (b) Draw the logic circuit for the given logic statement:

$$X = (A \text{ XOR } B) \text{ AND } (B \text{ OR NOT } C)$$

Do **not** attempt to simplify the logic statement. All logic gates must have a maximum of **two** inputs.



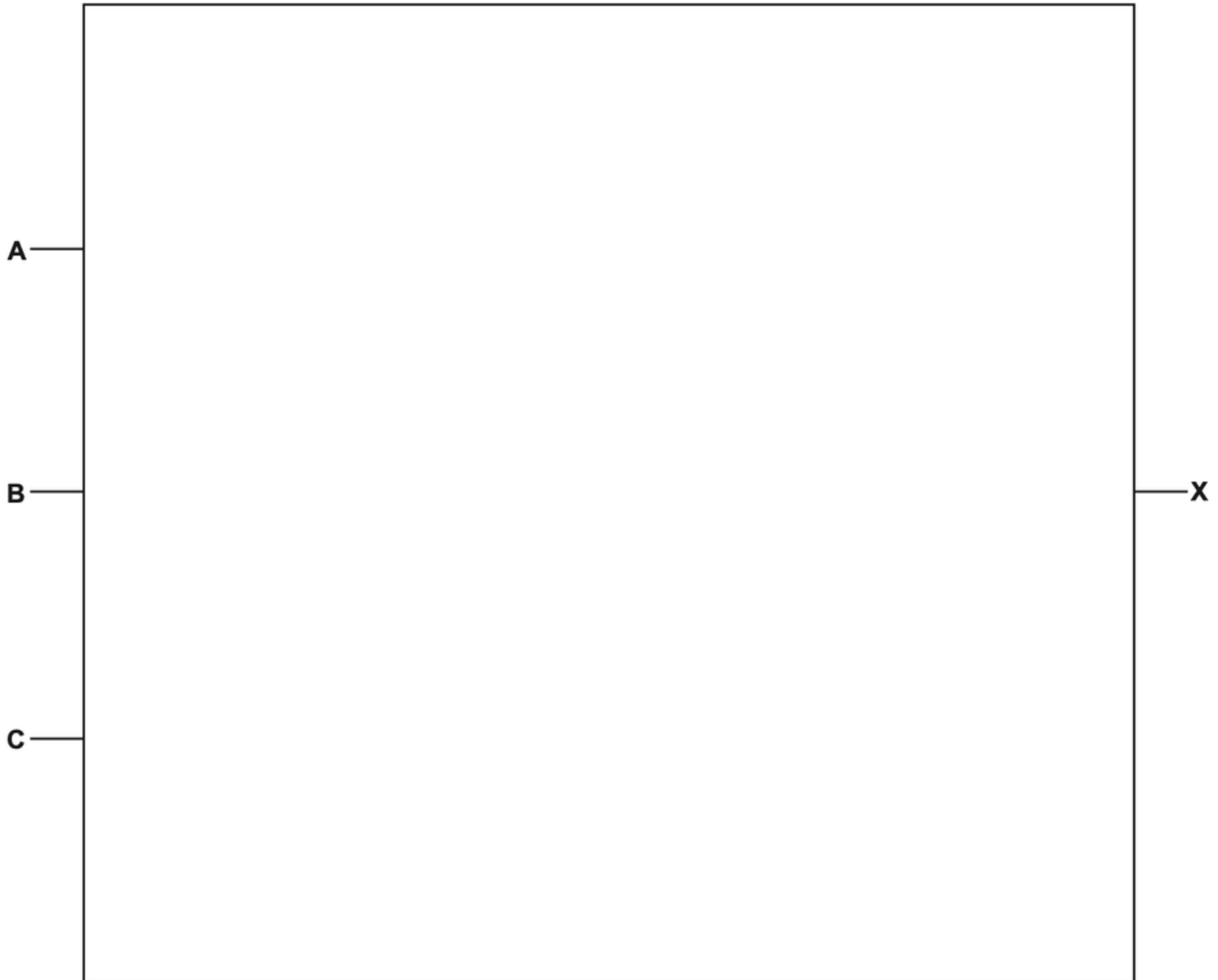
[4]

8 Consider the following logic statement:

$$X = (((A \text{ AND NOT } B) \text{ OR } (\text{NOT } (B \text{ NOR } C)))) \text{ AND } C$$

(a) Draw a logic circuit to represent the given logic statement.

Do **not** attempt to simplify the logic statement. All logic gates must have a maximum of **two** inputs.



(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

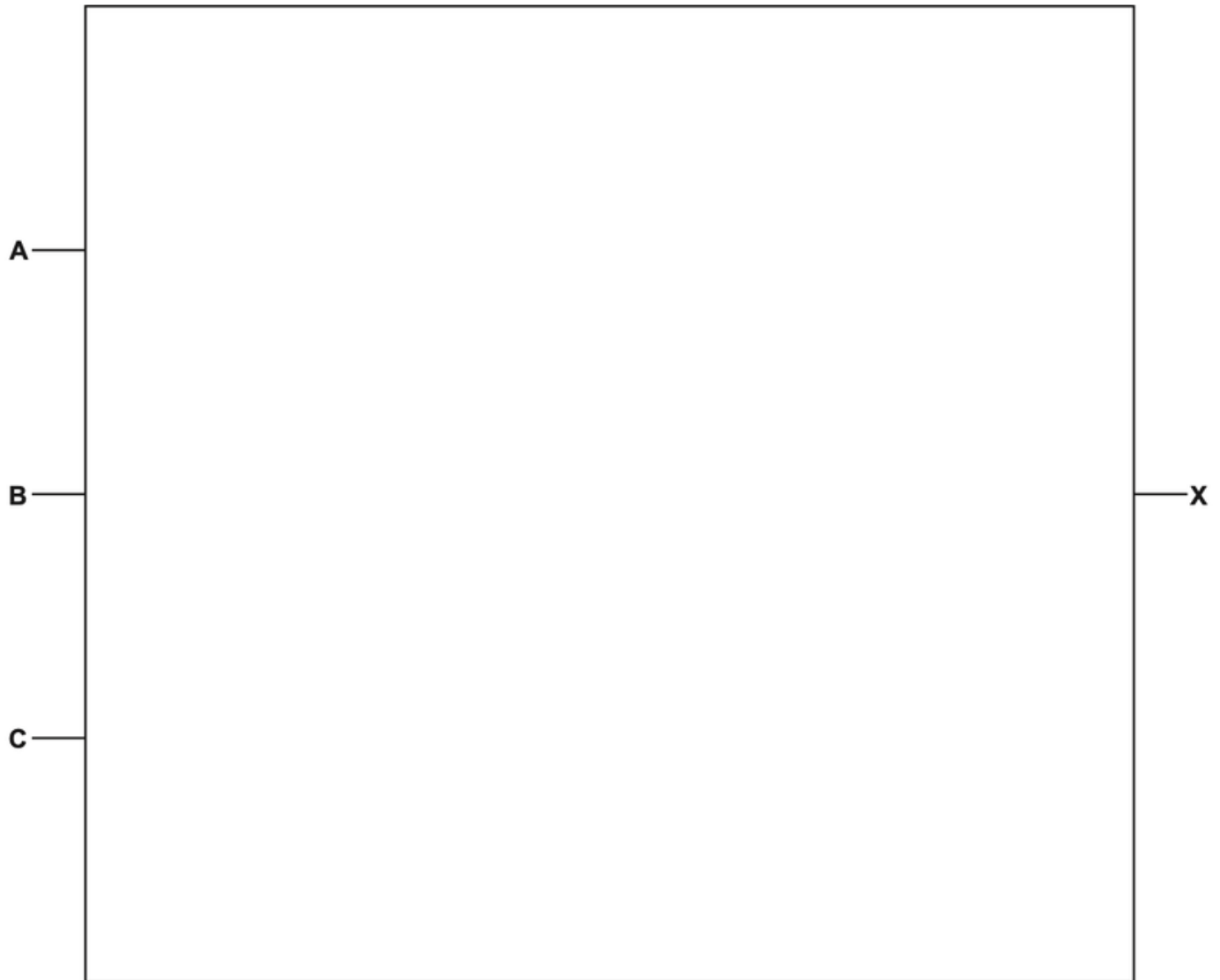
[4]

5 Consider the following logic statement:

$$X = ((A \text{ OR } B) \text{ AND } (\text{NOT } (B \text{ XOR } C))) \text{ AND } C$$

(a) Draw a logic circuit to represent the given logic statement.

Do **not** attempt to simplify the logic statement. All logic gates must have a maximum of **two** inputs.



[5]

(b) Complete the truth table for the given logic statement.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

3 AND, OR and XOR are three examples of logic gates.

(a) Four statements are given about these logic gates.

Tick (✓) to show which statements apply to each logic gate. Some statements may apply to more than **one** logic gate.

Statement	AND (✓)	OR (✓)	XOR (✓)
if both inputs are 0, the output is 0			
if both inputs are different, the output is 1			
if both inputs are 1, the output is 1			
if both inputs are the same, the output is always 0			

[4]

(b) NOT, AND, OR and XOR are all examples of logic gates.

State the name of **two** other logic gates and complete a truth table for each.

Logic gate 1

Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

Logic gate 2

Truth table:

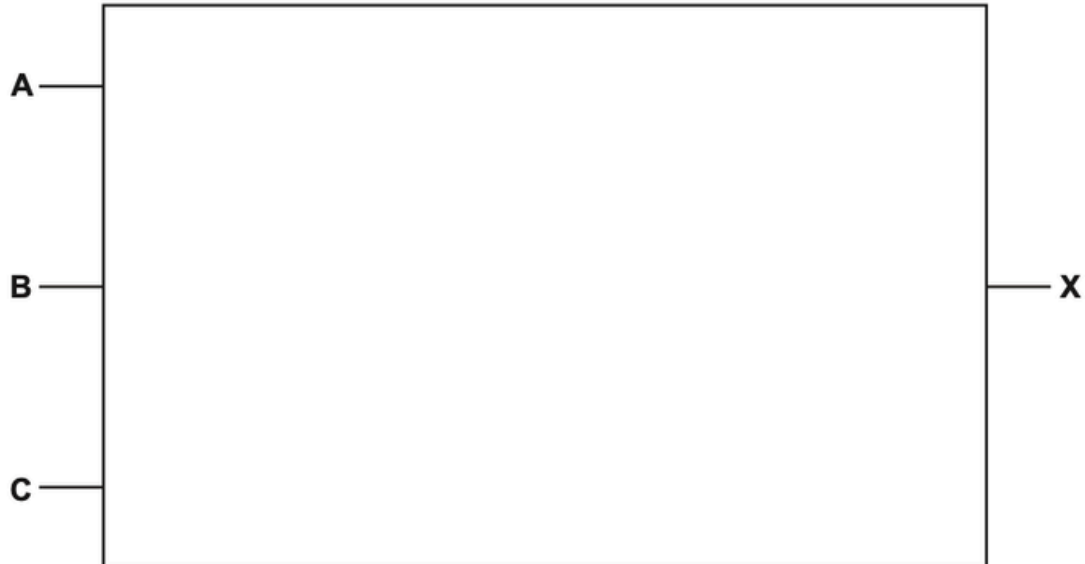
A	B	Output
0	0	
0	1	
1	0	
1	1	

[4]

7 Consider this logic expression.

$$X = (A \text{ OR NOT } B) \text{ AND } (B \text{ AND NOT } C)$$

- (a) Draw a logic circuit for this logic expression. Each logic gate must have a maximum of **two** inputs. Do **not** simplify this logic expression.



[5]

- (b) Complete the truth table from the given logic expression.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

9 Consider this logic expression.

$$Z = (\text{NOT } A \text{ OR } B) \text{ AND } (B \text{ XOR } C)$$

(a) Draw a logic circuit for this logic expression.

Each logic gate must have a maximum of **two** inputs.

Do **not** simplify this logic expression.



[4]

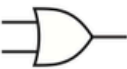
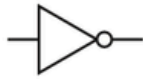
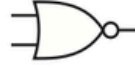
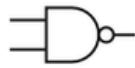
(b) Complete the truth table from the given logic expression.

A	B	C	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

2 Four logic gates and five standard symbols for logic gates are shown.

Draw **one** line to link each logic gate to its standard symbol. **Not** all standard symbols will be used.

Logic gate	Standard symbol
AND	
OR	
NAND	
NOT	
	

[4]

8 Consider this logic expression.

$$X = (A \text{ OR } B) \text{ AND } (\text{NOT } B \text{ AND } C)$$

Complete the truth table for this logic expression.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

8 Consider the logic expression:

Z is 1 if (**A** = 1 AND **C** = NOT 1) AND (**B** = 1 NOR **C** = 1)

(a) Draw a logic circuit for this logic expression.

Each logic gate must have a maximum of **two** inputs.

Do **not** simplify this logic expression.



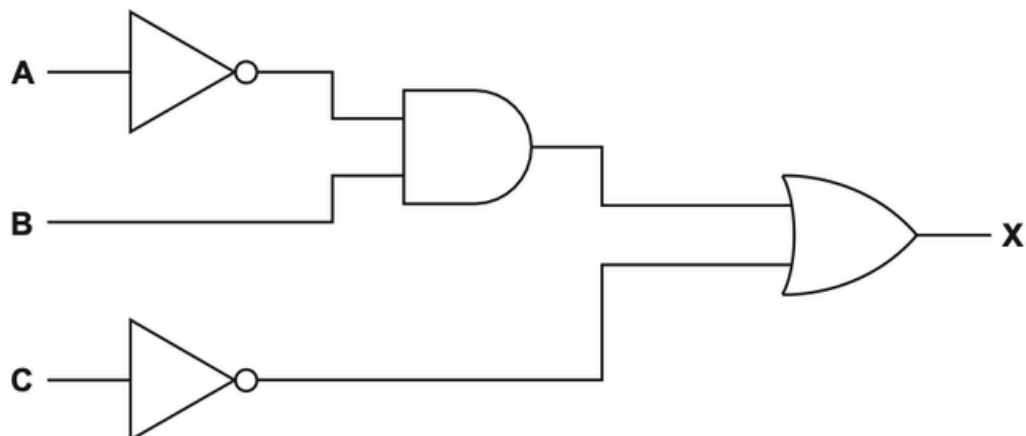
[4]

(b) Complete the truth table from the given logic expression.

A	B	C	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

7 Consider this logic circuit.



(a) Write a logic expression for this logic circuit. Do **not** attempt to simplify this logic expression.

$X =$

..... [4]

(b) Complete the truth table from the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

9 Consider the logic expression:

$$Z = (A \text{ NAND } B) \text{ OR NOT } (B \text{ XOR } C)$$

(a) Draw a logic circuit for this logic expression.

Each logic gate must have a maximum of **two** inputs.

Do **not** simplify this logic expression.



[4]

(b) Complete the truth table from the given logic expression.

A	B	C	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

- 6 There are **three** descriptions of logic gates. Each logic gate has two inputs **A** and **B** with one output **X**.

Identify each logic gate.

Complete a truth table for each logic gate.

- (a) The only time the output is 1 is when both inputs are 1.

Logic gate

Complete the truth table for this description.

A	B	X
0	0	
0	1	
1	0	
1	1	

[2]

- (b) The output is 1 when both inputs are different.

Logic gate

Complete the truth table for this description.

A	B	X
0	0	
0	1	
1	0	
1	1	

[2]

- (c) The only time the output is 1 is when both inputs are 0.

Logic gate

Complete the truth table for this description.

A	B	X
0	0	
0	1	
1	0	
1	1	

[2]

(d) Consider this logic expression:

$$X = (\text{NOT } A \text{ OR NOT } B) \text{ OR NOT } C$$

Draw a logic circuit for this logic expression. Each logic gate must have a maximum of **two** inputs. Do **not** attempt to simplify this logic expression.



[5]